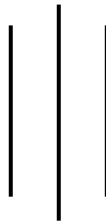




Tribhuvan University

Faculty of Humanities and Social Sciences



Network Administration Lab Report

Submitted To

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*In partial fulfilment of the requirements for the degree of bachelor's in
computer application.*

Submitted By

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Lab 1: Installation and Configuration of Linux and Windows Server

Objectives

To install Ubuntu Server and Windows Server in a virtualized environment and configure hostname, users, disk partitions and network settings.

Theory

A server operating system provides shared network services such as authentication, web hosting, file sharing, DHCP, DNS and remote administration. Installing the server correctly is the first step in building a reliable enterprise network.

Virtualization tools such as VirtualBox and VMware allow several operating systems to run on a single physical laptop. This makes it possible to practice server administration safely without affecting the host machine.

Network Diagram / Topology

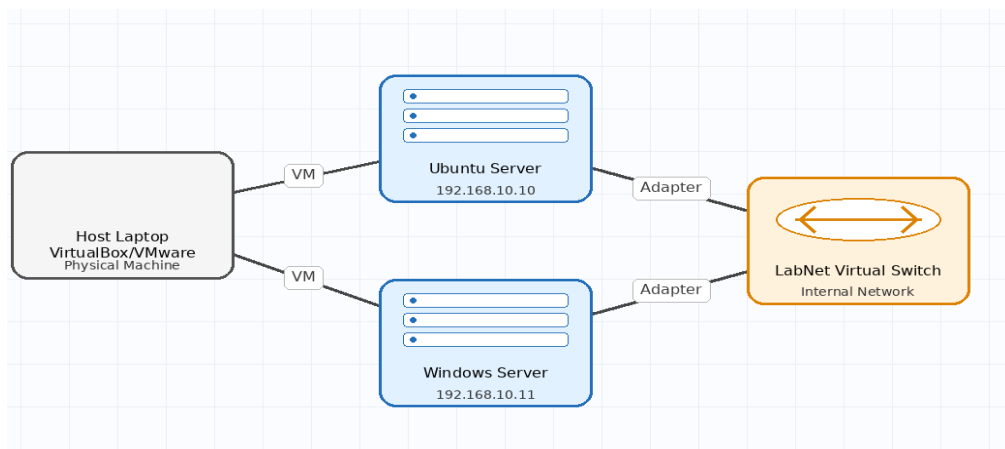


Figure 1.1: Network topology for Installation and Configuration of Linux and Windows Server

Tools Required

- VirtualBox or VMware Workstation
- Ubuntu Server 22.04/24.04 or CentOS Stream ISO
- Windows Server 2019/2022 ISO
- Minimum 8 GB RAM on host machine
- Internet connection for updates

Addressing / Configuration Plan

Machine	IP Address	Purpose
Ubuntu Server	192.168.10.10/24	Linux server
Windows Server	192.168.10.11/24	Windows server
Gateway	192.168.10.1	Default route
DNS	8.8.8.8 / local DNS	Name resolution

Step-by-Step Procedure

1. Open VirtualBox/VMware and create two VMs: one for Ubuntu/CentOS and one for Windows Server.
2. Assign at least 2 GB RAM, 2 CPU cores and 25 GB disk to each server VM.
3. Mount the ISO image and start the installation.
4. During installation, create an administrator user and set a strong password.
5. Configure disk partitions using automatic guided partitioning or manual partitions for /, /home and swap.
6. After installation, set the hostname and configure static network settings.
7. Restart the network service and verify the IP address.
8. Install guest additions/tools if required and take a clean baseline snapshot.

Important Commands / Configuration

```
# Ubuntu hostname and user configuration
sudo hostnamectl set-hostname ubuntu-server
sudo adduser netadmin
sudo usermod -aG sudo netadmin
lsblk
ip addr show
sudo nano /etc/netplan/00-installer-config.yaml
sudo netplan apply
ping -c 4 8.8.8.8
```

```
# Windows Server PowerShell
Rename-Computer -NewName WIN-SERVER -Restart
New-LocalUser "labuser" -Password (Read-Host -AsSecureString "Password")
Add-LocalGroupMember -Group "Administrators" -Member "labuser"
New-NetIPAddress -InterfaceAlias "Ethernet" -IPAddress 192.168.10.11 -PrefixLength 24 -
DefaultGateway 192.168.10.1
Set-DnsClientServerAddress -InterfaceAlias "Ethernet" -ServerAddresses 8.8.8.8
```

Screenshots / Practical Evidence

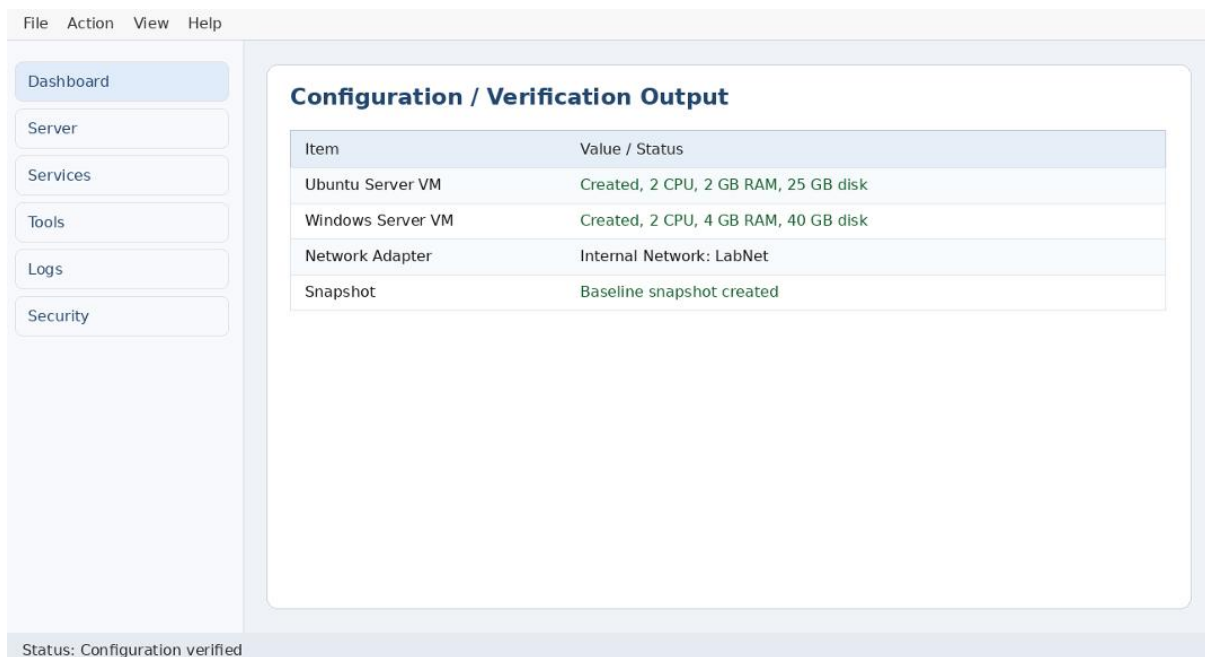


Figure 1.2: VirtualBox Manager - Ubuntu and Windows Server VMs

Observations

Test	Observation
Ubuntu installation	Completed without error
Windows Server installation	Completed and administrator account created
Static IP configuration	IP address remained fixed after reboot
Connectivity	Ping test successful between servers

Result

Ubuntu/CentOS and Windows Server were successfully installed and configured in a virtual environment.

Conclusion

This practical developed the foundation for server administration by installing and configuring Linux and Windows Server VMs with proper identity, users, storage and networking.

Lab 2: Design and Implementation of a Virtual Network

Objectives

To create a virtual network with one server and two client machines, assign IP addresses and verify end-to-end connectivity.

Theory

A virtual network connects multiple virtual machines using virtual adapters and virtual switches. The most common modes are NAT, bridged, host-only and internal network.

In an internal network, VMs can communicate with each other but remain isolated from the outside network. This is useful for laboratory environments because experiments do not disturb the real network.

Network Diagram / Topology

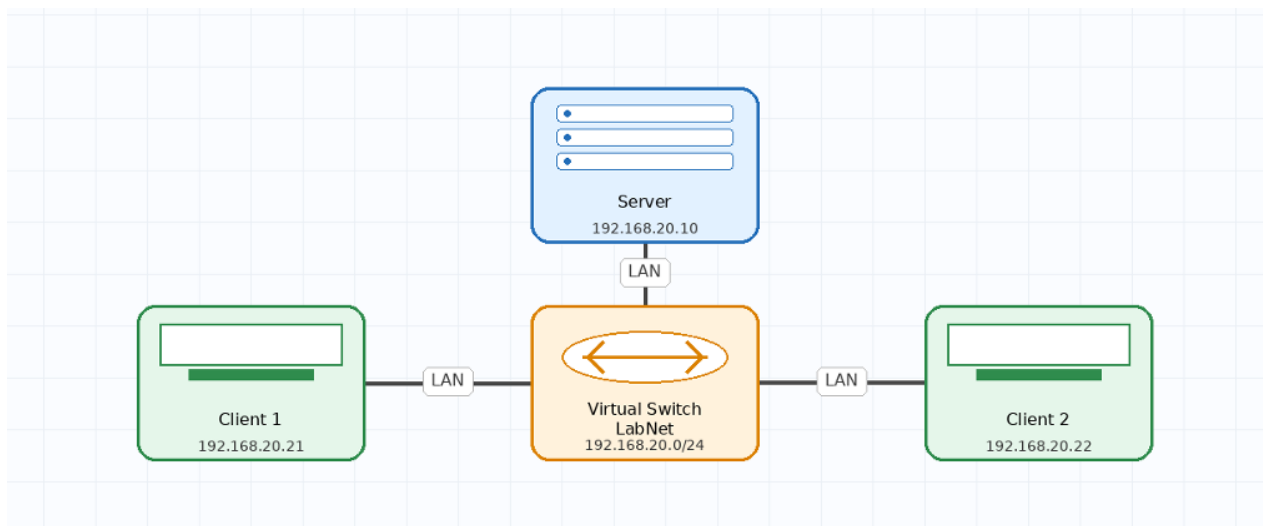


Figure 2.1: Network topology for Design and Implementation of a Virtual Network

Tools Required

- VirtualBox/VMware
- Ubuntu Server VM
- Windows or Linux Client 1
- Windows or Linux Client 2
- Command Prompt/Terminal

Addressing / Configuration Plan

Machine	IP Address	Subnet Mask
Server	192.168.20.10	255.255.255.0
Client 1	192.168.20.21	255.255.255.0
Client 2	192.168.20.22	255.255.255.0
Network	192.168.20.0/24	Lab LAN

Step-by-Step Procedure

1. Create three VMs: Server, Client1 and Client2.
2. Set all VM adapters to the same internal network named LabNet.
3. Boot each VM and configure static IP addresses from the addressing table.
4. Disable firewall temporarily if ICMP is blocked in the lab.
5. Ping the server from both clients.
6. Ping Client1 from Client2 to confirm peer connectivity.
7. Document the successful replies and latency values.

Important Commands / Configuration

```
# Linux static IP temporary configuration
sudo ip addr add 192.168.20.10/24 dev enp0s3
sudo ip link set enp0s3 up
ip addr show enp0s3

# Windows client configuration
netsh interface ip set address name="Ethernet" static 192.168.20.21 255.255.255.0

# Connectivity test
ping 192.168.20.10
ping 192.168.20.22
arp -a
```

Screenshots / Practical Evidence

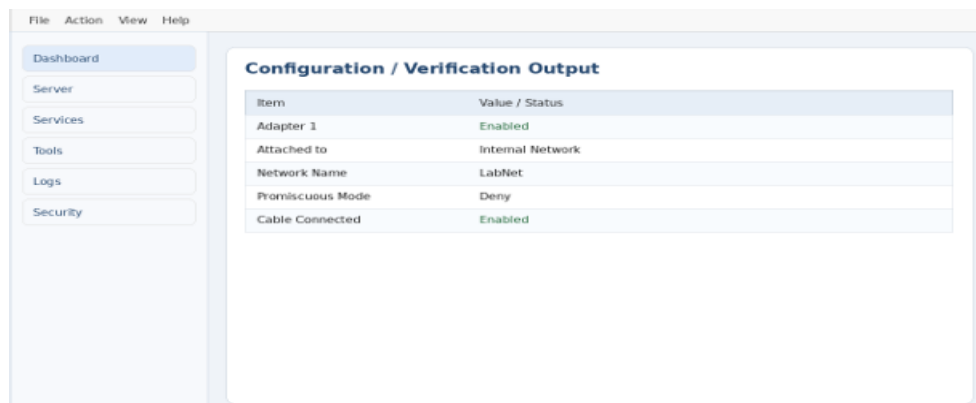


Figure 2.2: VirtualBox Internal Network Settings

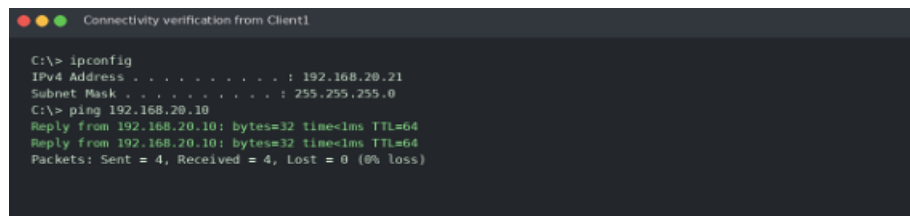


Figure 2.3: Connectivity verification from Client1

Observations

Test	Observation
Internal network mode	All VMs attached to LabNet
Server to client ping	Successful
Client to client ping	Successful
ARP table	MAC addresses learned successfully

Result

A virtual network with one server and two clients was successfully designed, configured and tested.

Conclusion

The practical demonstrated how virtual switches and adapters can be used to build a small working LAN on a single laptop.

Lab 3: IP Addressing, Subnetting, and Network Troubleshooting

Objectives

To design subnets, configure static IP addresses and troubleshoot network connectivity using common diagnostic commands.

Theory

Subnetting is the process of dividing a larger network into smaller logical networks. It improves address management, security and traffic control.

Each subnet has a network address, usable host range and broadcast address. A host can communicate directly only with hosts in the same subnet unless a router is available.

Troubleshooting involves checking the physical/virtual adapter, IP address, subnet mask, default gateway, DNS settings, routes and listening ports.

Network Diagram / Topology

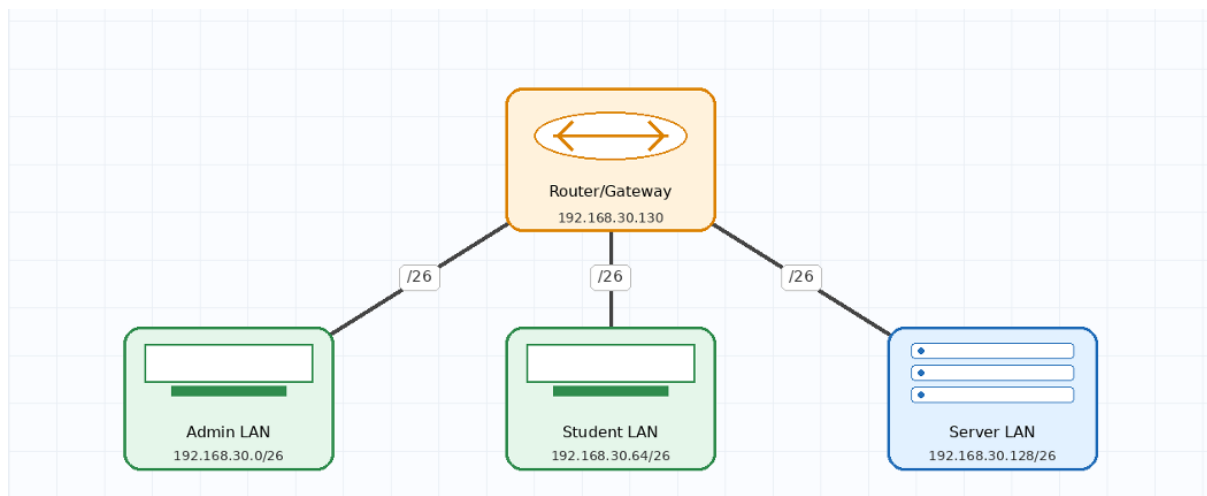


Figure 3.1: Network topology for IP Addressing, Subnetting, and Network Troubleshooting

Tools Required

- Linux Terminal
- Windows Command Prompt/PowerShell
- ipcalc or subnet calculator
- ping/traceroute/tracert
- netstat/ss

Addressing / Configuration Plan

Subnet	Network	Usable Range	Broadcast
Admin LAN	192.168.30.0/26	192.168.30.1-192.168.30.62	192.168.30.63
Student LAN	192.168.30.64/26	192.168.30.65-192.168.30.126	192.168.30.127
Server LAN	192.168.30.128/26	192.168.30.129-192.168.30.190	192.168.30.191

Step-by-Step Procedure

1. Choose a base network such as 192.168.30.0/24.
2. Divide it into /26 subnets to support up to 62 usable hosts per subnet.
3. Assign gateway and host addresses according to the subnet table.
4. Configure static IP addresses on the systems.
5. Verify the IP address, subnet mask and routing table.
6. Use ping to test local connectivity.
7. Use traceroute/tracert to identify route path.
8. Use netstat or ss to check active connections and listening services.

Important Commands / Configuration

```
# Subnet verification
ipcalc 192.168.30.0/26

# Linux IP configuration
sudo ip addr add 192.168.30.129/26 dev enp0s3
sudo ip route add default via 192.168.30.130
ip route
ping -c 4 192.168.30.130
traceroute 8.8.8.8
ss -tulnp

# Windows commands
ipconfig /all
route print
ping 192.168.30.129
tracert 8.8.8.8
netstat -ano
```

Screenshots / Practical Evidence

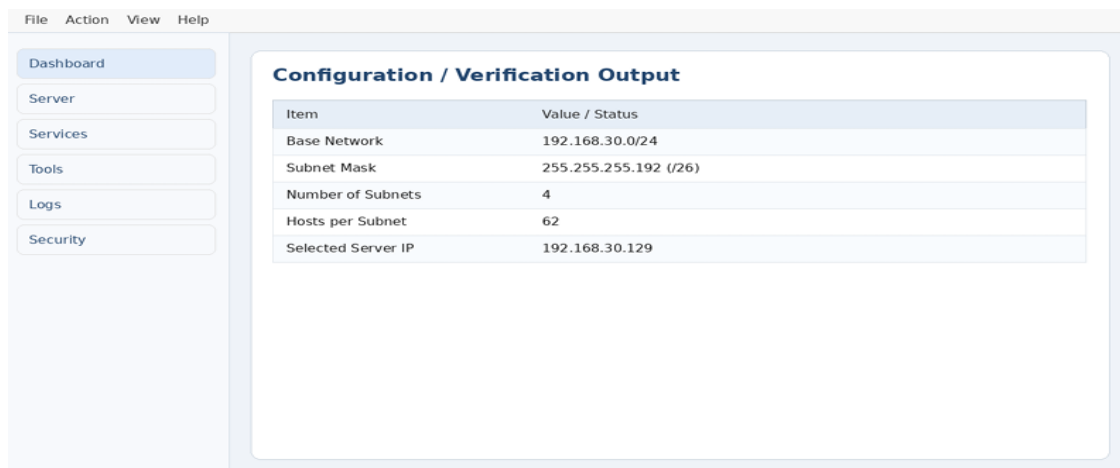


Figure 3.2: Subnet Planning Worksheet

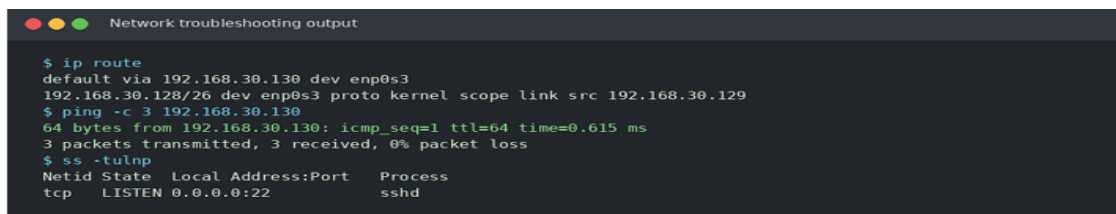


Figure 3.3: Network troubleshooting output

Observations

Test	Observation
Subnet calculation	Network, host and broadcast addresses identified
Static IP	Configured correctly
Default route	Displayed correctly in route table
Troubleshooting commands	Helped locate connectivity status

Result

Subnetting, static IP addressing and network troubleshooting commands were successfully practiced.

Conclusion

This practical improved understanding of IPv4 planning and systematic troubleshooting of network connectivity problems.

Lab 4: Installation and Configuration of DHCP and DNS Services

Objectives

To configure DHCP for automatic IP assignment and DNS for forward and reverse name resolution.

Theory

DHCP automatically assigns IP addresses and related settings such as subnet mask, gateway and DNS server to client devices. It reduces manual configuration errors and simplifies network administration.

DNS translates names into IP addresses and IP addresses back into names. Forward lookup resolves names to IP addresses, while reverse lookup resolves IP addresses to names.

Network Diagram / Topology

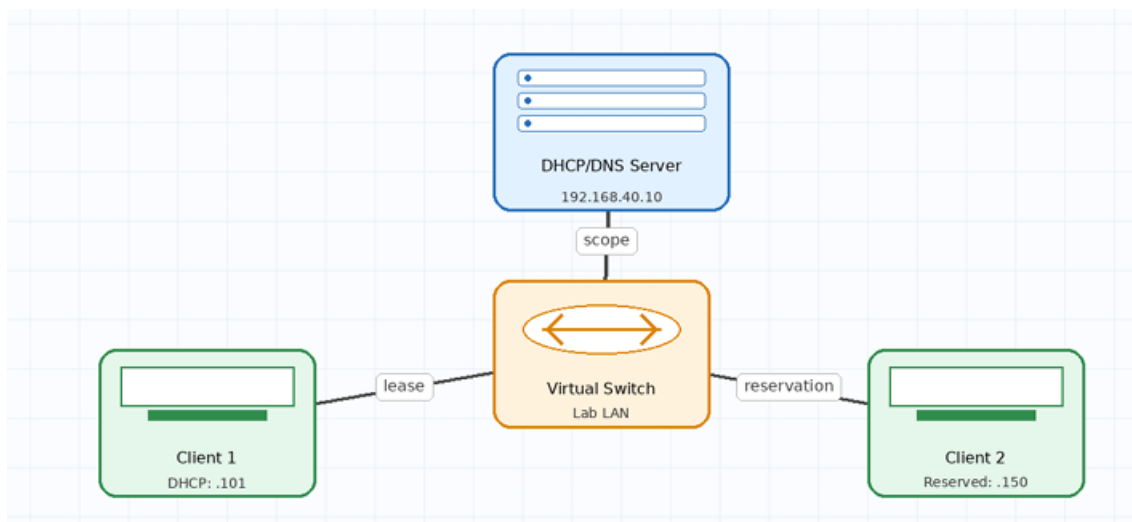


Figure 4.1: Network topology for Installation and Configuration of DHCP and DNS Services

Tools Required

- Windows Server DHCP/DNS roles or Ubuntu isc-dhcp-server and BIND9
- Windows/Linux clients
- nslookup/dig
- ipconfig/dhclient

Addressing / Configuration Plan

Service	Configuration
DHCP Scope	192.168.40.100 - 192.168.40.200
Gateway Option	192.168.40.1
DNS Server Option	192.168.40.10
Domain Name	lab.local
DNS A Record	server.lab.local -> 192.168.40.10

Step-by-Step Procedure

1. Install DHCP and DNS roles on Windows Server or equivalent Linux packages.
2. Configure a static IP address on the server.
3. Create a DHCP scope with start IP, end IP, subnet mask and lease duration.
4. Add DHCP options for gateway, DNS server and domain name.
5. Create a DHCP reservation using the client MAC address.
6. Create a forward lookup zone named lab.local.
7. Add records for important hosts.
8. Create reverse lookup zone and PTR records.
9. Renew client IP addresses and test DNS resolution.

Important Commands / Configuration

Windows Server PowerShell

```
Install-WindowsFeature DHCP,DNS -IncludeManagementTools
```

```
Add-DhcpServerv4Scope -Name "LabScope" -StartRange 192.168.40.100 -EndRange 192.168.40.200 -  
SubnetMask 255.255.255.0
```

```
Set-DhcpServerv4OptionValue -DnsServer 192.168.40.10 -Router 192.168.40.1 -DnsDomain lab.local
```

```
Add-DhcpServerv4Reservation -ScopeId 192.168.40.0 -IPAddress 192.168.40.150 -ClientId "00-11-22-33-  
44-55" -Description "Client1"
```

```
Add-DnsServerPrimaryZone -Name "lab.local" -ZoneFile "lab.local.dns"
```

```
Add-DnsServerResourceRecordA -Name "server" -ZoneName "lab.local" -IPv4Address 192.168.40.10
```

```
Add-DnsServerPrimaryZone -NetworkId "192.168.40.0/24" -ReplicationScope "Domain"
```

Client verification

```
ipconfig /release
```

```
ipconfig /renew
```

```
ipconfig /all
```

```
nslookup server.lab.local
```

```
nslookup 192.168.40.10
```

Screenshots / Practical Evidence

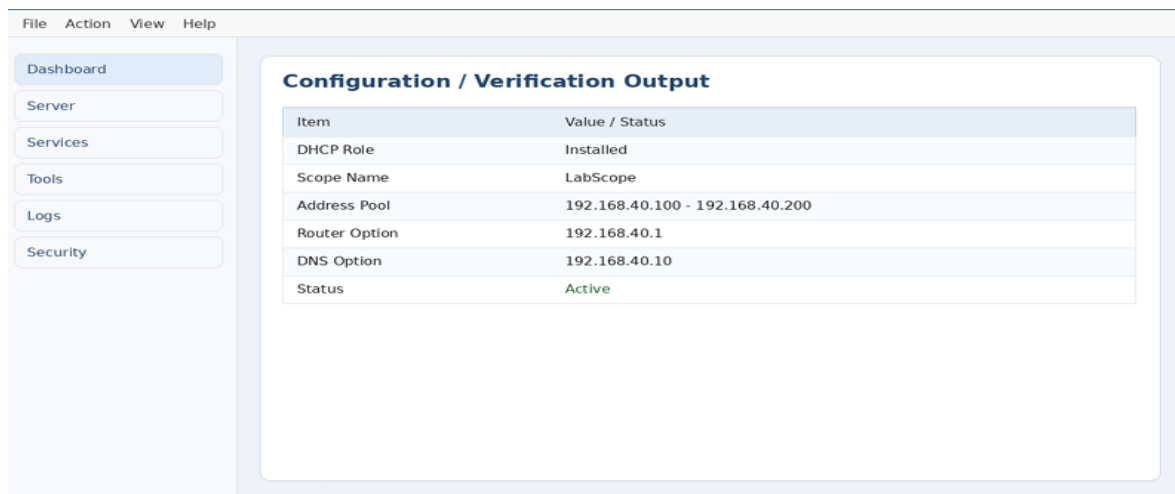


Figure 4.2: Windows Server DHCP Scope Configuration

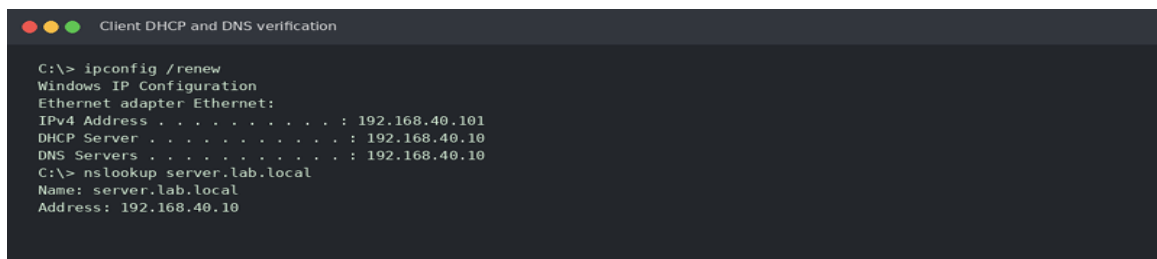


Figure 4.3: Client DHCP and DNS verification

Observations

Test	Observation
DHCP scope	Activated successfully
Client lease	Client received IP automatically
Forward lookup	server.lab.local resolved correctly
Reverse lookup	IP resolved back to hostname

Result

DHCP and DNS services were installed, configured and verified successfully.

Conclusion

This practical showed how centralized IP allocation and name resolution simplify network management.

Lab 5: Centralized User Authentication using Active Directory/OpenLDAP

Objectives

To configure centralized authentication using Active Directory or OpenLDAP, create users/groups, join a client to the domain and apply policies.

Theory

A directory service stores information about users, groups, computers and resources. It enables centralized authentication and authorization across an enterprise network.

Microsoft Active Directory uses domains, domain controllers, DNS, organizational units and Group Policy. OpenLDAP provides similar directory-based authentication using LDAP entries and schemas.

Network Diagram / Topology

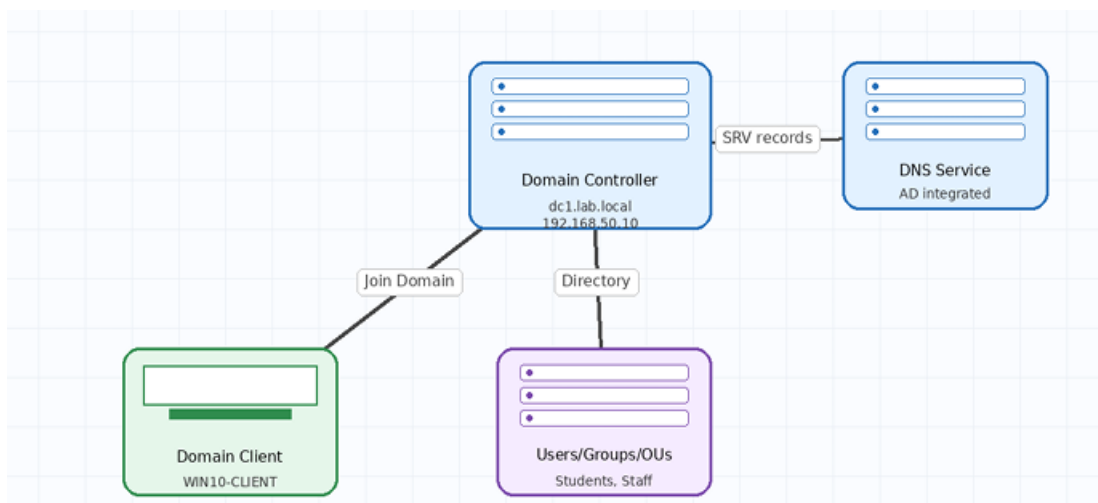


Figure 5.1: Network topology for Centralized User Authentication using Active Directory/OpenLDAP

Tools Required

- Windows Server Active Directory Domain Services or OpenLDAP
- Windows Client or Linux Client
- DNS service
- Group Policy Management Console / ldap-utils

Addressing / Configuration Plan

Component	Value
Domain	lab.local
Domain Controller	dc1.lab.local / 192.168.50.10
OU	Students, Staff, Servers
Group	IT_Admins, Students
Client	WIN10-CLIENT / 192.168.50.21

Step-by-Step Procedure

1. Assign a static IP address to the server.
2. Install Active Directory Domain Services and DNS.
3. Promote the server as a domain controller for lab.local.
4. Create organizational units for Students, Staff and Servers.
5. Create user accounts and security groups.
6. Add users to appropriate groups.
7. Join the client computer to the lab.local domain.
8. Log in from the client using domain user credentials.
9. Apply a simple Group Policy such as password policy or desktop restriction.

Important Commands / Configuration

```
# Windows Server AD DS installation
Install-WindowsFeature AD-Domain-Services -IncludeManagementTools
Install-ADDSForest -DomainName "lab.local" -DomainNetbiosName "LAB"
New-ADOrganizationalUnit -Name "Students" -Path "DC=lab,DC=local"
New-ADUser -Name "Ramesh Student" -SamAccountName ramesh -AccountPassword (Read-Host -
AsSecureString "Password") -Enabled $true
New-ADGroup -Name "Students" -GroupScope Global
Add-ADGroupMember -Identity "Students" -Members ramesh

# Client domain join
Add-Computer -DomainName lab.local -Credential LAB\Administrator -Restart
gpupdate /force
gpresult /r

# OpenLDAP basic verification
ldapsearch -x -LLL -b dc=lab,dc=local
```

Screenshots / Practical Evidence

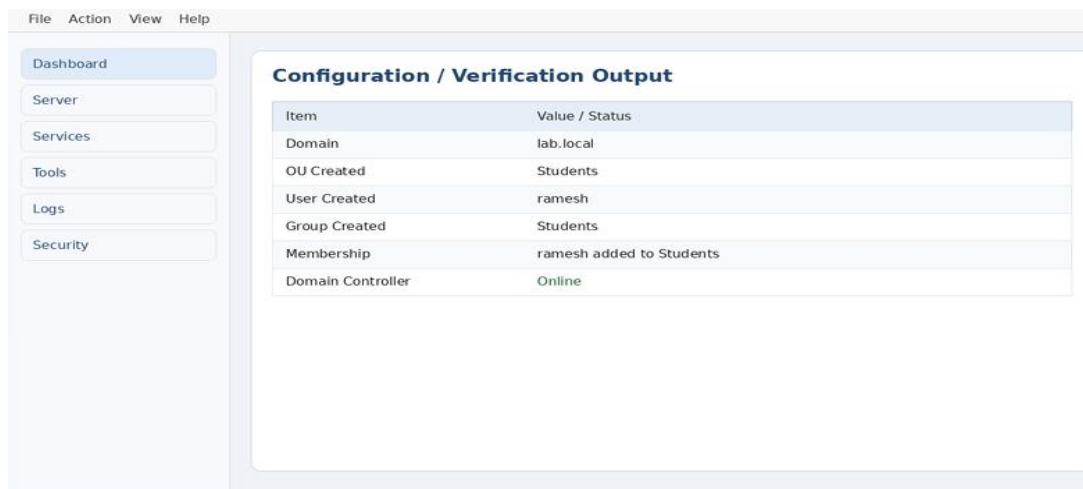


Figure 5.2: Active Directory Users and Computers

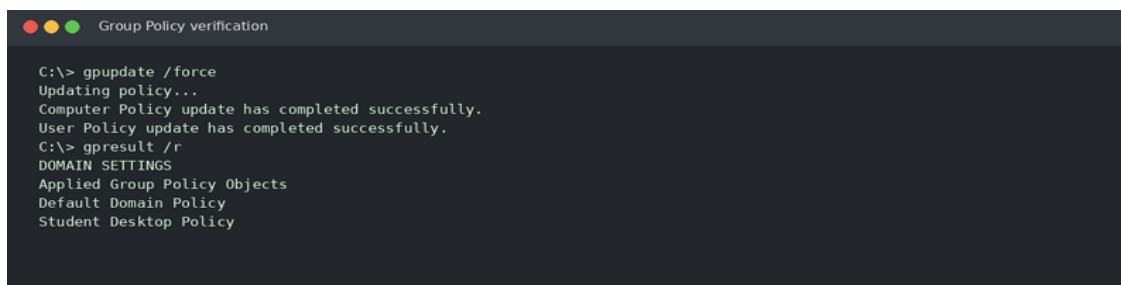


Figure 5.3: Group Policy verification

Observations

Test	Observation
Domain creation	lab.local created successfully
Users/groups	Created and organized in OUs
Client join	Client joined domain successfully
Policy application	Group Policy applied after gpupdate

Result

Centralized authentication was successfully configured and tested using domain login.

Conclusion

This practical demonstrated the importance of directory services in centralized identity, authentication and policy management.

Lab 6: Configuration of Remote Access Services (SSH and FTP)

Objectives

To configure DHCP for automatic IP assignment and DNS for forward and reverse name resolution.

Theory

Remote access services allow administrators to manage servers without being physically present. SSH provides encrypted command-line access and secure file transfer.

FTP transfers files but traditional FTP is not encrypted. In secure environments, SFTP or FTPS is preferred. Still, FTP is useful for understanding service configuration and access permissions.

Network Diagram / Topology

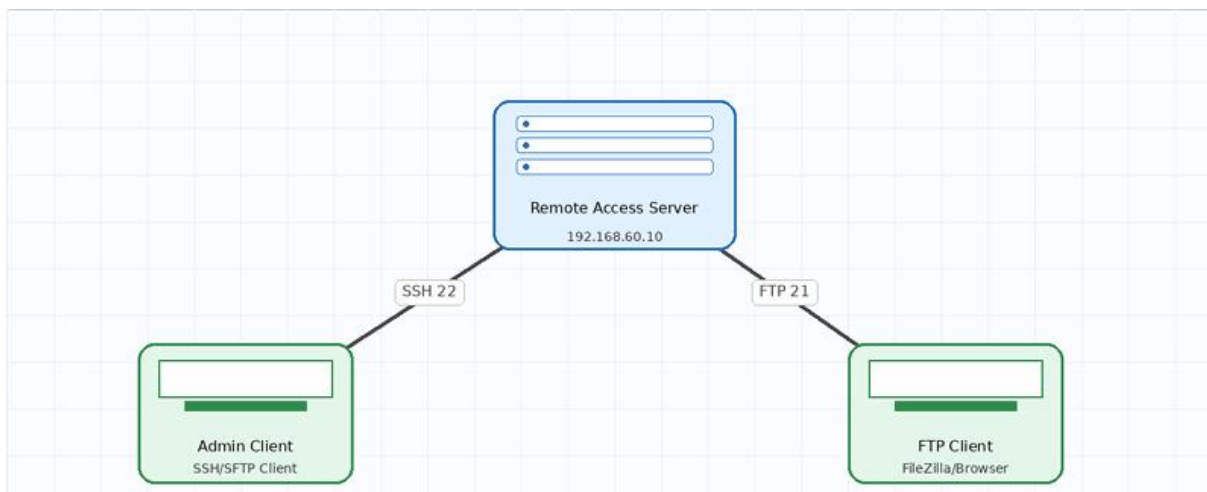


Figure 6.1: Network topology for Configuration of Remote Access Services (SSH and FTP)

Tools Required

- Ubuntu Server
- OpenSSH Server
- vsftpd or FileZilla Server
- PuTTY/Terminal
- WinSCP/FileZilla Client
- UFW firewall

Addressing / Configuration Plan

Service	Port	Security Note
SSH	22/TCP	Encrypted remote shell
SFTP	22/TCP	Secure file transfer over SSH
FTP	21/TCP	Use only in lab or with FTPS
Server IP	192.168.60.10	Remote access host

Step-by-Step Procedure

1. Install openssh-server and verify that ssh service is running.
2. Create a normal user for remote login.
3. Connect from client using ssh username@server-ip.
4. Transfer a test file using scp or sftp.
5. Install vsftpd and enable local user login.
6. Create an FTP directory and assign ownership/permissions.
7. Configure chroot_local_user for user isolation.
8. Allow required ports in the firewall.
9. Disable root login in SSH and restart services.

Important Commands / Configuration

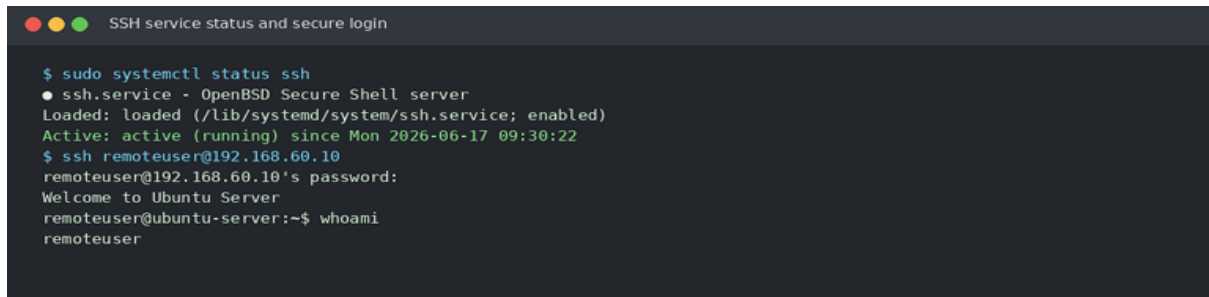
```
sudo apt update
sudo apt install openssh-server vsftpd -y
sudo systemctl enable --now ssh
sudo systemctl status ssh
sudo adduser remoteuser
ssh remoteuser@192.168.60.10
scp report.txt remoteuser@192.168.60.10:/home/remoteuser/
sftp remoteuser@192.168.60.10
```

```
sudo nano /etc/ssh/sshd_config
PermitRootLogin no
PasswordAuthentication yes
sudo systemctl restart ssh
```

```
sudo nano /etc/vsftpd.conf
local_enable=YES
write_enable=YES
chroot local user=YES
```

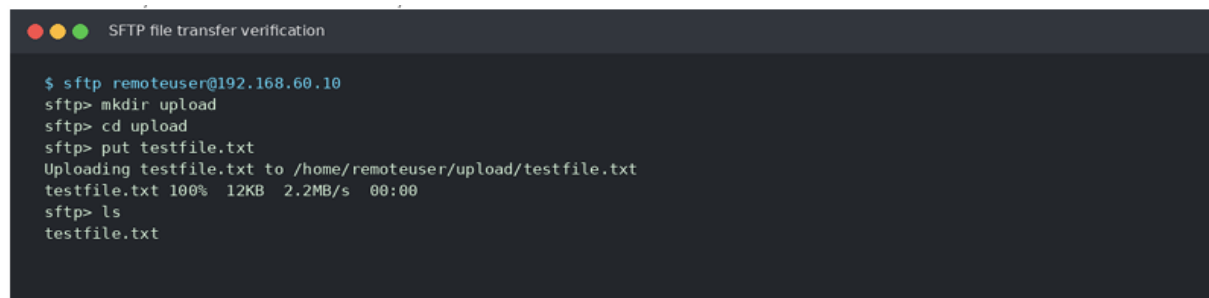
```
sudo systemctl restart vsftpd
sudo ufw allow 22/tcp
sudo ufw allow 21/tcp
```

Screenshots / Practical Evidence



```
$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/lib/systemd/system/ssh.service; enabled)
   Active: active (running) since Mon 2026-06-17 09:30:22
$ ssh remoteuser@192.168.60.10
remoteuser@192.168.60.10's password:
Welcome to Ubuntu Server
remoteuser@ubuntu-server:~$ whoami
remoteuser
```

Figure 6.2: SSH service status and secure login



```
$ sftp remoteuser@192.168.60.10
sftp> mkdir upload
sftp> cd upload
sftp> put testfile.txt
Uploading testfile.txt to /home/remoteuser/upload/testfile.txt
testfile.txt 100% 12KB 2.2MB/s 00:00
sftp> ls
testfile.txt
```

Figure 6.3: SFTP file transfer verification

Observations

Test	Observation
SSH login	Successful using remote user
SFTP upload	File uploaded successfully
FTP access	User could access assigned directory
Hardening	Root login disabled and firewall configured

Result

SSH and FTP services were successfully installed, tested and hardened with basic security settings.

Conclusion

This lab showed how remote administration and file transfer services are deployed and secured in a server environment.

Lab 7: Installation and Configuration of Web Services

Objectives

To configure Apache/Nginx and IIS web servers, host websites, create virtual hosts and analyze server logs.

Theory

A web server delivers web pages to clients using HTTP or HTTPS. Apache and Nginx are common Linux web servers, while IIS is Microsoft's web server platform.

Virtual hosting allows one server to host multiple websites using different hostnames, IP addresses or ports. This is widely used in enterprise and hosting environments.

Web server logs record requests, response codes, client IP addresses and errors. Administrators use logs for troubleshooting, auditing and performance analysis.

Network Diagram / Topology

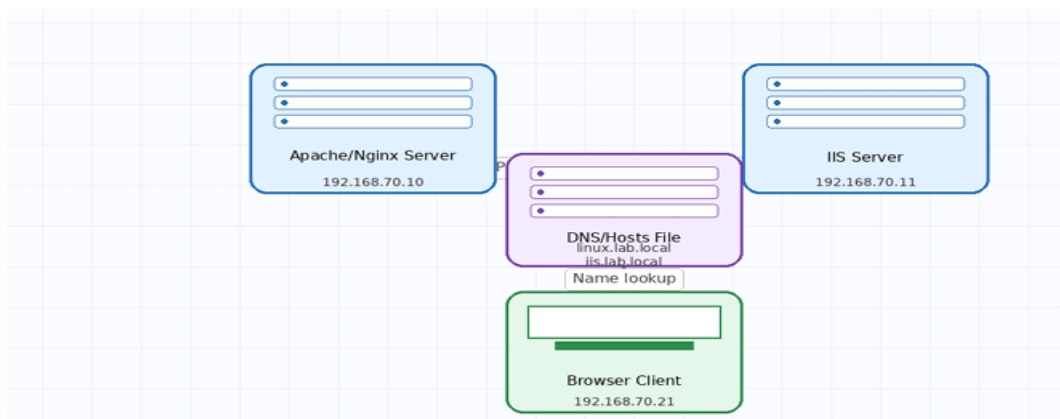


Figure 7.1: Network topology for Installation and Configuration of Web Services

Tools Required

- Ubuntu Server
- Apache2 or Nginx
- Windows Server IIS
- Browser
- curl
- Log files

Addressing / Configuration Plan

Website	Server	URL
Linux Apache Site	192.168.70.10	http://linux.lab.local
Windows IIS Site	192.168.70.11	http://iis.lab.local
Virtual Host Root	/var/www/linux.lab.local	DocumentRoot
Apache Logs	/var/log/apache2/access.log	Access log

Step-by-Step Procedure

1. Install Apache or Nginx on Ubuntu Server.
2. Create a website directory and sample index.html page.
3. Create a virtual host configuration file.
4. Enable the site and reload the web server.
5. Install IIS role on Windows Server.
6. Create a new IIS website and bind it to a hostname or port.
7. Add DNS/hosts file entries for test domain names.
8. Access the websites from a client browser.
9. Analyze access and error logs after generating traffic.

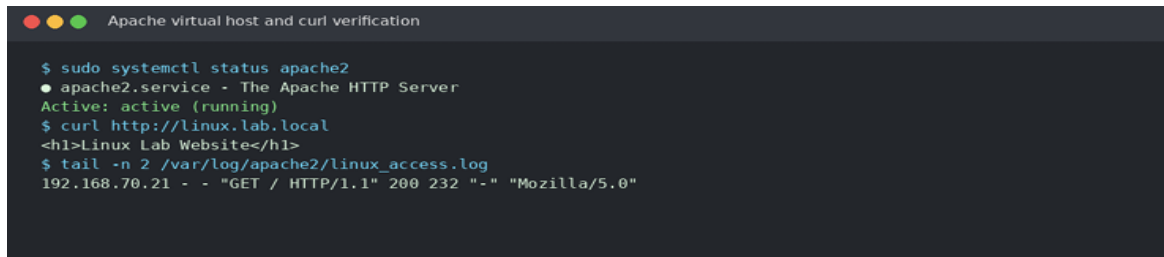
Important Commands / Configuration

```
# Apache installation and virtual host
sudo apt update
sudo apt install apache2 -y
sudo mkdir -p /var/www/linux.lab.local
echo "<h1>Linux Lab Website</h1>" | sudo tee /var/www/linux.lab.local/index.html
sudo nano /etc/apache2/sites-available/linux.lab.local.conf
<VirtualHost *:80>
    ServerName linux.lab.local
    DocumentRoot /var/www/linux.lab.local
    ErrorLog ${APACHE_LOG_DIR}/linux_error.log
    CustomLog ${APACHE_LOG_DIR}/linux_access.log combined
</VirtualHost>
sudo a2ensite linux.lab.local.conf
sudo systemctl reload apache2
curl http://linux.lab.local
tail -f /var/log/apache2/linux_access.log
```

```
# IIS PowerShell
```

```
Install-WindowsFeature Web-Server -IncludeManagementTools
New-Item -Path C:\inetpub\iis.lab.local -ItemType Directory
New-Website -Name "IISLab" -Port 80 -HostHeader "iis.lab.local" -PhysicalPath "C:\inetpub\iis.lab.local"
```

Screenshots / Practical Evidence



```
$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
Active: active (running)
$ curl http://linux.lab.local
<h1>Linux Lab Website</h1>
$ tail -n 2 /var/log/apache2/linux_access.log
192.168.70.21 - - "GET / HTTP/1.1" 200 232 "-" "Mozilla/5.0"
```

Figure 7.2: Apache virtual host and curl verification

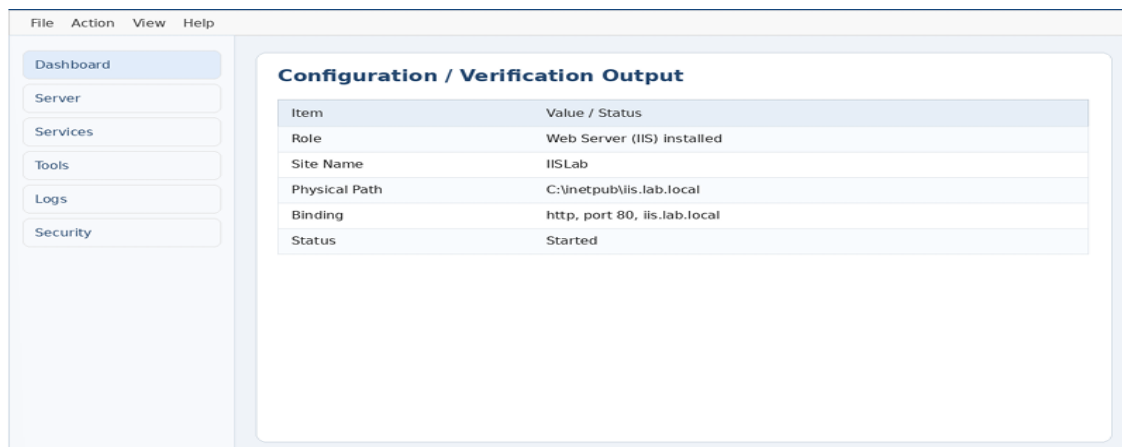


Figure 7.3: IIS Manager - Website Binding

Observations

Test	Observation
Apache service	Running
Linux website	Loaded successfully in browser
IIS site	Started and accessible
Logs	Access logs recorded client requests

Result

Apache/Nginx and IIS web services were configured and sample websites were successfully hosted.

Conclusion

This lab provided hands-on experience in hosting websites and managing Linux and Windows web server platforms.

Lab 8: Centralized Data Storage using Samba and NFS

Objectives

To configure centralized file sharing between Linux and Windows systems using Samba and NFS with different access permissions.

Theory

Centralized storage allows users to store files on a server instead of individual workstations. This simplifies backup, permission management and collaboration.

Samba implements SMB/CIFS file sharing and allows Linux servers to share files with Windows clients. NFS is commonly used for file sharing between Unix/Linux systems.

Network Diagram / Topology

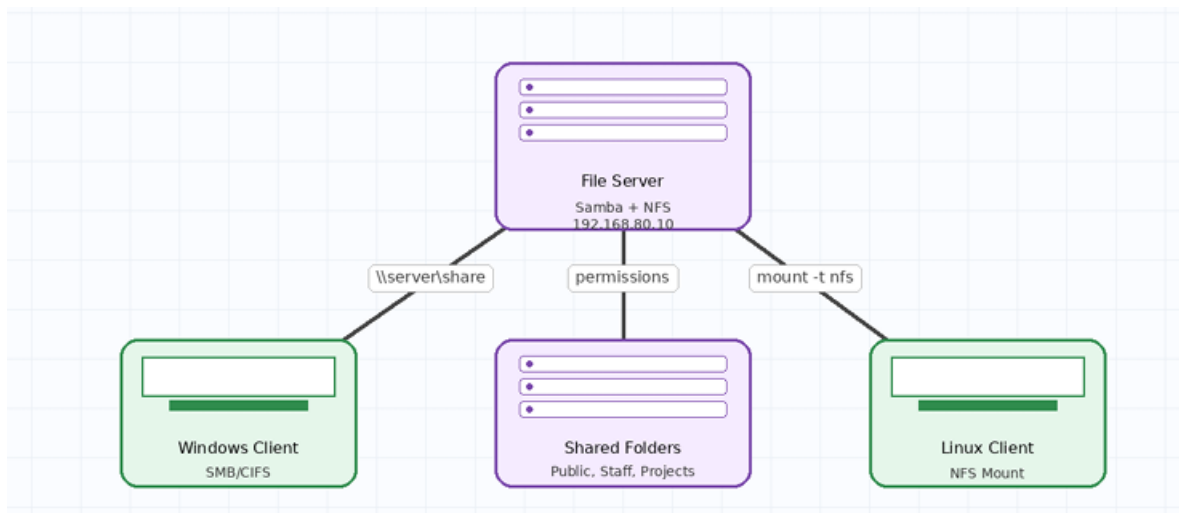


Figure 8.1: Network topology for Centralized Data Storage using Samba and NFS

Tools Required

- Ubuntu Server
- Samba
- NFS Kernel Server
- Windows Client
- Linux Client
- mount/cifs-utils

Addressing / Configuration Plan

Share	Protocol	Access
/srv/samba/public	Samba	Read-only for guests
/srv/samba/staff	Samba	Read-write for staff group
/srv/nfs/projects	NFS	Read-write for Linux clients
Server IP	192.168.80.10	File server

Step-by-Step Procedure

1. Install Samba and NFS server packages.
2. Create shared directories under /srv.
3. Create Linux users and groups for access control.
4. Set ownership and permissions on share folders.
5. Edit smb.conf and add public and staff shares.
6. Create Samba passwords for users.
7. Restart Samba services and test from Windows using \\server-ip\share.
8. Configure NFS exports in /etc/exports.
9. Run exportfs and mount NFS share from Linux client.
10. Verify read-only and read-write behavior.

Important Commands / Configuration

```
sudo apt install samba nfs-kernel-server cifs-utils -y
sudo groupadd staff
sudo useradd -m -G staff staffuser
sudo smbpasswd -a staffuser
sudo mkdir -p /srv/samba/public /srv/samba/staff /srv/nfs/projects
sudo chown -R root:staff /srv/samba/staff
sudo chmod -R 2770 /srv/samba/staff
```

```
# /etc/samba/smb.conf
[Public]
path = /srv/samba/public
browseable = yes
read only = yes
guest ok = yes
```

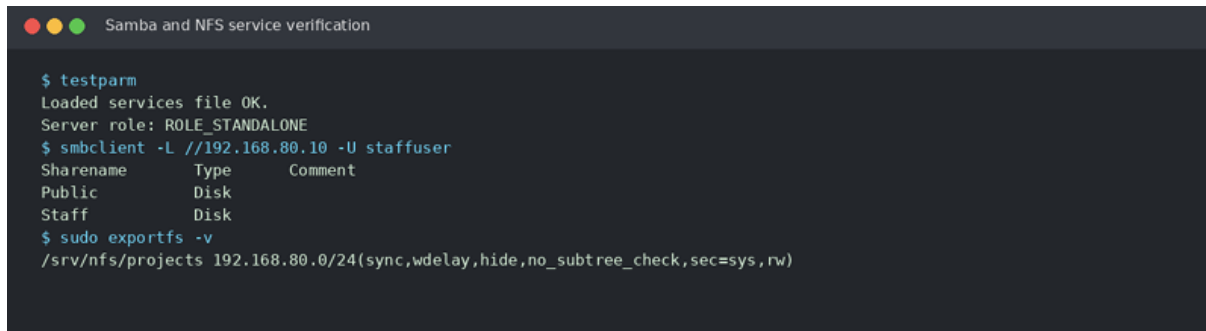
```
[Staff]
path = /srv/samba/staff
```

```
valid users = @staff
read only = no
browseable = yes
```

```
sudo systemctl restart smbd nmbd
```

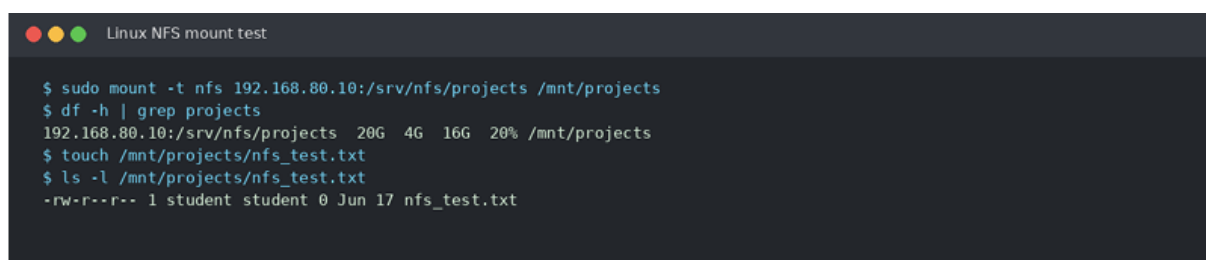
```
# /etc/exports
/srv/nfs/projects 192.168.80.0/24(rw,sync,no_subtree_check)
sudo exportfs -ra
sudo mount -t nfs 192.168.80.10:/srv/nfs/projects /mnt
```

Screenshots / Practical Evidence



```
$ testparm
Loaded services file OK.
Server role: ROLE_STANDALONE
$ smbclient -L //192.168.80.10 -U staffuser
Sharename      Type      Comment
Public         Disk
Staff          Disk
$ sudo exportfs -v
/srv/nfs/projects 192.168.80.0/24(sync,wdelay,hide,no_subtree_check,sec=sys,rw)
```

Figure 8.2: Samba and NFS service verification



```
$ sudo mount -t nfs 192.168.80.10:/srv/nfs/projects /mnt/projects
$ df -h | grep projects
192.168.80.10:/srv/nfs/projects 20G 4G 16G 20% /mnt/projects
$ touch /mnt/projects/nfs_test.txt
$ ls -l /mnt/projects/nfs_test.txt
-rw-r--r-- 1 student student 0 Jun 17 nfs_test.txt
```

Figure 8.3: Linux NFS mount test

Observations

Test	Observation
Samba public share	Accessible as read-only
Samba staff share	Writable only by authorized user
NFS export	Mounted successfully on Linux client
Permissions	Access control worked as configured

Result

Centralized file sharing was successfully configured using Samba and NFS.

Conclusion

This practical demonstrated cross-platform data sharing and access control using Samba and NFS.

Lab 9: Installation and Configuration of Email Services

Objectives

To configure Postfix and Dovecot, create user mailboxes and send/receive emails using Thunderbird. Objectives:

Theory

Email systems use several protocols. SMTP is used to send mail, while IMAP and POP3 are used to retrieve mail from a server.

In a lab environment, local domain mail can be tested without public DNS. For real-world deployment, MX records, TLS certificates, spam protection and authentication policies are required.

Network Diagram / Topology

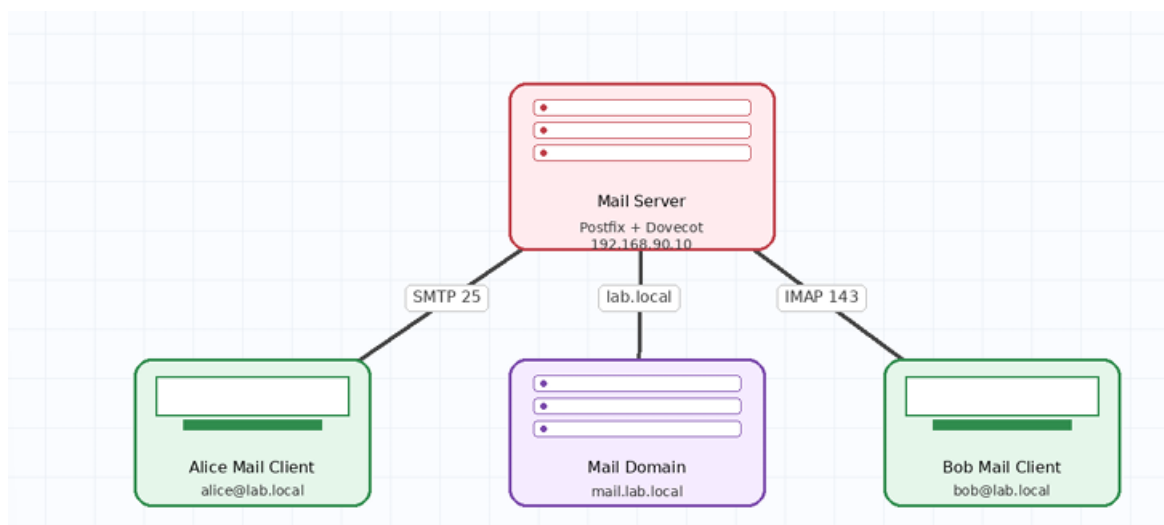


Figure 9.1: Network topology for Installation and Configuration of Email Services

Tools Required

- Ubuntu Server
- Postfix
- Dovecot IMAP/POP3
- Thunderbird mail client
- telnet or swaks for testing

Addressing / Configuration Plan

Component	Value
Mail Domain	lab.local
Mail Server	mail.lab.local / 192.168.90.10
SMTP Port	25
IMAP Port	143
Users	alice, bob

Step-by-Step Procedure

1. Set hostname as mail.lab.local.
2. Install Postfix and choose Internet Site or local mail configuration.
3. Configure myhostname, mydomain and mydestination.
4. Install Dovecot IMAP/POP3 packages.
5. Set mailbox format to Maildir if required.
6. Create two local users such as alice and bob.
7. Restart Postfix and Dovecot services.
8. Configure Thunderbird with SMTP and IMAP server settings.
9. Send a message from alice to bob and verify receiving.

Important Commands / Configuration

```
sudo hostnamectl set-hostname mail.lab.local
sudo apt update
sudo apt install postfix dovecot-imapd dovecot-pop3d mailutils -y
sudo postconf -e "myhostname = mail.lab.local"
sudo postconf -e "mydomain = lab.local"
sudo postconf -e "mydestination = $myhostname, localhost.$mydomain, localhost, $mydomain"
sudo postconf -e "home_mailbox = Maildir/"
sudo systemctl restart postfix dovecot
sudo adduser alice
sudo adduser bob

# Send local test mail
echo "Hello Bob" | mail -s "Lab Test Mail" bob
sudo tail -f /var/log/mail.log

# Thunderbird settings
Incoming: IMAP, Server: 192.168.90.10, Port: 143

Outgoing: SMTP, Server: 192.168.90.10, Port: 25
```

Screenshots / Practical Evidence

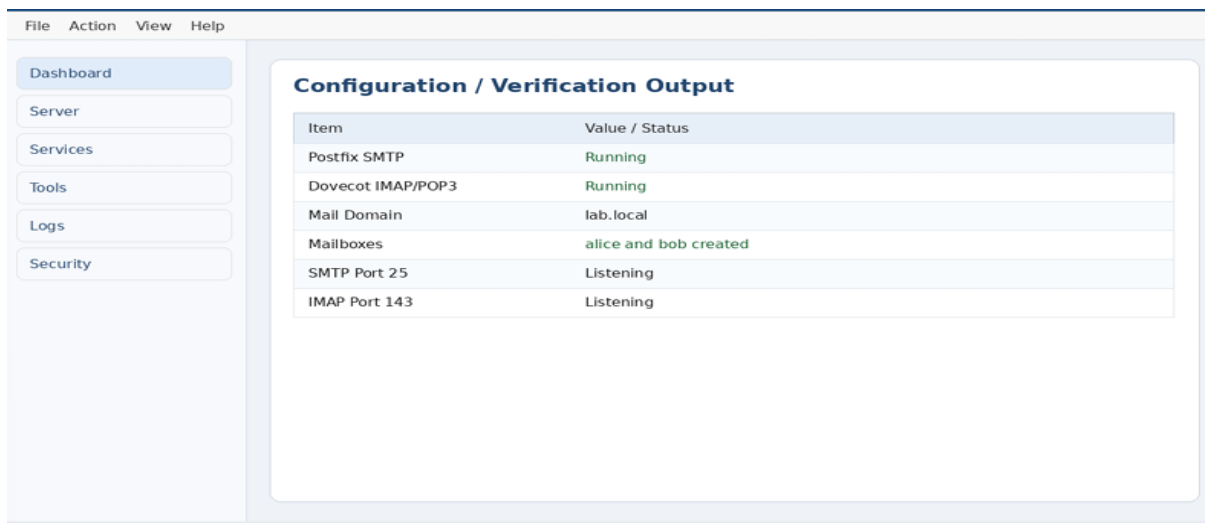


Figure 9.2: Postfix and Dovecot Service Status

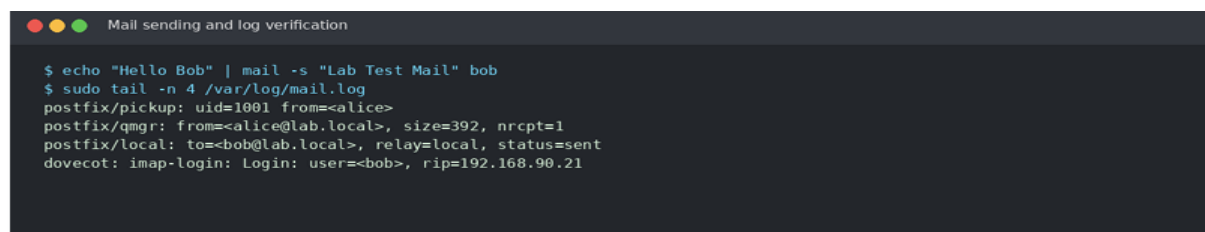


Figure 9.3: Mail sending and log verification

Observations

Test	Observation
SMTP service	Postfix running and accepting local mail
IMAP service	Dovecot allowed mailbox login
User mailbox	Message delivered to bob
Thunderbird	Sent and received test message successfully

Result

Postfix and Dovecot email services were configured and tested with Thunderbird.

Conclusion

This lab explained the working of a basic mail server and demonstrated SMTP/IMAP mail flow in a lab network.

Lab 10: Service Monitoring and Network Traffic Analysis

Objectives

To monitor server services using Nagios/Zabbix and analyze HTTP, DNS and ICMP traffic using Wireshark.

Theory

Service monitoring helps administrators know whether important services are running, overloaded or unavailable. Tools such as Nagios and Zabbix provide dashboards, alerts and historical metrics.

Network traffic analysis shows what packets are moving across the network. Wireshark captures frames and decodes protocols such as ICMP, DNS, HTTP, TCP and UDP.

Monitoring and packet analysis together support troubleshooting, performance tuning and security investigation.

Network Diagram / Topology

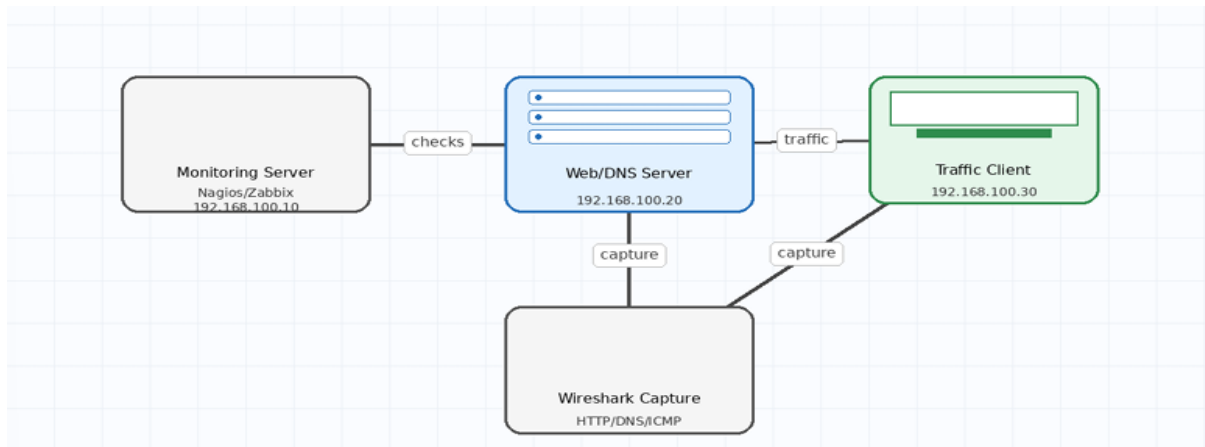


Figure 10.1: Network topology for Service Monitoring and Network Traffic Analysis

Tools Required

- Nagios Core or Zabbix
- Wireshark
- Ubuntu/Windows clients
- Apache/DNS services
- ping/curl/nslookup

Addressing / Configuration Plan

Component	IP Address	Purpose
Monitoring Server	192.168.100.10	Nagios/Zabbix
Web/DNS Server	192.168.100.20	Monitored host
Client	192.168.100.30	Traffic generator
Protocols	ICMP, DNS, HTTP	Analysis

Step-by-Step Procedure

Install Nagios/Zabbix server or use a prebuilt monitoring appliance.

1. Add the web/DNS server as a monitored host.
2. Configure checks for ping, HTTP and DNS services.
3. Verify dashboard status and service alerts.
4. Open Wireshark on the client or monitoring machine.
5. Start capture on the correct interface.
6. Generate ICMP traffic using ping.
7. Generate DNS traffic using nslookup.
8. Generate HTTP traffic using curl or browser.
9. Apply display filters and analyze protocol details.

Important Commands / Configuration

```
# Generate traffic
ping 192.168.100.20
nslookup server.lab.local 192.168.100.20
curl http://192.168.100.20

# Wireshark display filters
icmp
dns
http
ip.addr == 192.168.100.20

# Nagios service checks examples
check_ping -H 192.168.100.20 -w 100.0,20% -c 500.0,60%
check_http -H 192.168.100.20
check_dns -H 192.168.100.20 -s 192.168.100.20
```

Screenshots / Practical Evidence

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000	192.168.10.20	192.168.10.1	DNS	74	Standard query A server.lab.local
2	0.002	192.168.10.1	192.168.10.20	DNS	90	Standard query response A 192.168.10.10
3	1.019	192.168.10.20	192.168.10.10	ICMP	98	Echo request id=0x04 sequence=1
4	1.021	192.168.10.10	192.168.10.20	ICMP	98	Echo reply id=0x04 sequence=1
5	2.313	192.168.10.20	192.168.10.10	HTTP	236	GET /index.html HTTP/1.1
6	2.320	192.168.10.10	192.168.10.20	HTTP	512	HTTP/1.1 200 OK (text/html)

Figure 10.2: Wireshark Packet Capture

```
Traffic generation commands

$ ping -c 2 192.168.100.20
64 bytes from 192.168.100.20: icmp_seq=1 ttl=64 time=0.8 ms
64 bytes from 192.168.100.20: icmp_seq=2 ttl=64 time=0.7 ms
$ nslookup server.lab.local 192.168.100.20
Name: server.lab.local
Address: 192.168.100.20
$ curl -I http://192.168.100.20
HTTP/1.1 200 OK
```

Figure 10.3: Traffic generation commands

Observations

Test	Observation
Monitoring dashboard	All configured services displayed OK
ICMP capture	Echo request and reply packets observed
DNS capture	Query and response packets observed
HTTP capture	GET request and 200 OK response observed

Result

Service monitoring and network packet analysis were successfully performed.

Conclusion

This practical showed how monitoring and packet analysis help maintain reliable network services.